SHOKS SAT

AUGUST PREDICTIONS

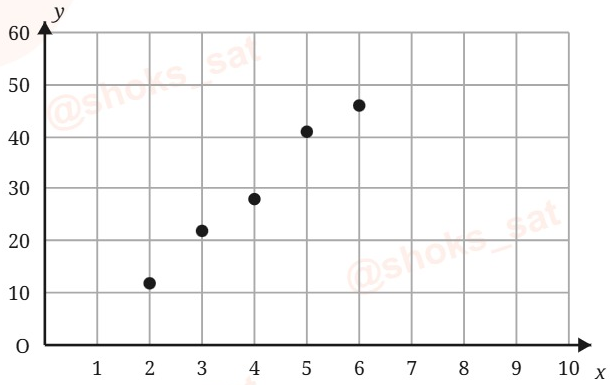
30 HARD SAT MATH QUESTIONS

SECTION ONE · QUESTIONS 1–13

MULTIPLE CHOICE

The scatterplot shows data set A, which consists of the weights y, in pounds, of a Labrador retriever puppy at various ages, x, in months. The equation of a line of best fit for the relationship in data set A can be written as $y = -5.0 + 8.7x$, where $2 \leq x \leq 6$.

The puppy was weighed again at 9 months old and weighed 52 pounds. Data set B consists of all the data points in data set A as well as the data point $(9, 52)$. The equation of a line of best fit for data set B can be written as $y = r + sx$, where r and s are constants and $2 \leq x \leq 9$. Assuming the equations of the lines of best fit are calculated in the same way, which of the following is the best estimate for the value of s?



A) 5.8

B) 8.7

C) 13.7

D) 17.7

A book-of-the-month club charges $17 for the first book purchased during the month and an additional $13 for each book after the first. Which equation describes this situation, where y is the total cost, in dollars, for books purchased during a month, x is the number of books purchased during the month, and $x > 0$?

A) $y = x + 17$

B) $y = 13x + 17$

C) $y = 13(x - 1) + 17$

D) $y = (x - 1) + 17$

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A company developed a plan to set the selling price of a product. The company determined that for a selling price of $120.00, zero products would be sold. For each $1.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly $1,966.50, which of the following could be the number of products sold?

(revenue = price × number of products sold)

A) 23

B) 40

C) 1,231

D) 2,400

To win a game show, a contestant needs to score at least 80 total points from two rounds. Correct responses in the first round are worth 4 points each, and correct responses in the second round are worth 5 points each. Which inequality models this situation, where f is the number of correct responses in the first round and s is the number of correct responses in the second round?

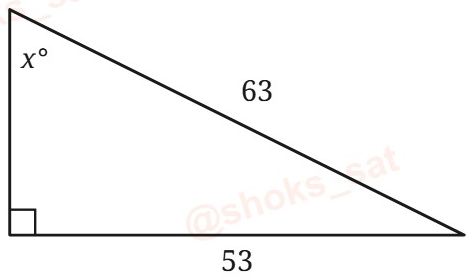
A) $f + s \leq 80$

B) $4f + 5s \geq 80$

C) $4f + 5s \leq 80$

D) $5f + 4s \geq 80$

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Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin x°$?

A) $\frac{116}{63}$

B) $\frac{53}{63}$

C) $\frac{10}{63}$

D) $\frac{1}{63}$

If $34 + 8(x + 3) = 7(x + 3) - 34$, what is the value of $x + 3$?

A) 68

B) 34

C) 0

D) -68

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$$x(x-18)=c$$

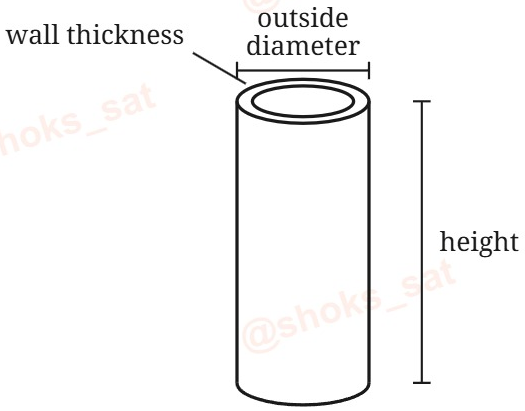
In the given equation, c is a constant. A solution to the equation is $k+7\sqrt{2}$, where k is an integer. What is the value of c?

A) 7

B) 17

C) 31

D) 98



Note: Figure not drawn to scale.

In the figure, a hollow steel pipe is in the shape of a right circular cylinder. The steel pipe has an outside diameter of 48 inches, a wall thickness of $\frac{3}{8}$ inches, and a height of 138 inches. Which of the following is closest to the volume, in cubic inches, of the wall of this steel pipe?

A) 61

B) 325

C) 1,242

D) 7,743

Which quadratic equation has no real solutions?

A) $x^2 + 24x - 144 = 0$

B) $x^2 - 24x + 144 = 0$

C) $7x^2 - 24x - 144 = 0$

D) $7x^2 - 24x + 144 = 0$

For the polynomial function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(-5, 0)$, $(4, 0)$, and $(9, 0)$. Which of the following must be a factor of $f(x)$?

A) $x + 4$

B) $x + 9$

C) $x - 4$

D) $x - 5$

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Circles A and B are graphed in the xy-plane. Circle A is represented by the equation $(x-9)^2+(y+9)^2=81$ and intersects the x-axis at the point (r, s). Circle B has its center at (r, s) and has a radius of the same length as circle A. Which equation represents circle B?

A) $x^2+(y-9)^2=81$

B) $x^2+(y+9)^2=81$

C) $(x-9)^2+y^2=81$

D) $(x+9)^2+y^2=81$

The function f is defined by the equation $f(x) = 6|x - 4| - 31$. What is the value of $f(-4)$?

A) -79

B) -31

C) 17

D) 79

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If $\frac{x-18}{36}=\frac{x-18}{9}$, what is the value of $x+18$?

A) 0

B) 18

C) 36

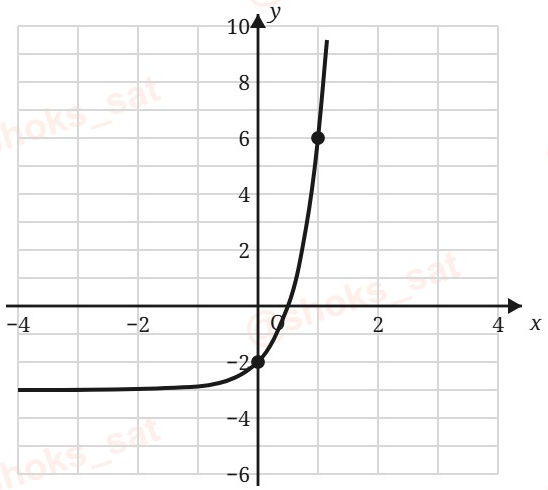
D) 54

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SECTION TWO · QUESTIONS 14–20

STUDENT-PRODUCED

No answer options. Enter your answer in the response box.



The equation of the graph shown is $y = a^x + b$, where a and b are constants. What is the value of $a - b$?

STUDENT-PRODUCED RESPONSE

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x - 650)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. What is the value of w?

STUDENT-PRODUCED RESPONSE

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$$-x^2 + bx - 49 = 0$$

In the given equation, b is an integer. The equation has no real solutions. What is the least possible value of b?

STUDENT-PRODUCED RESPONSE

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An entomologist placed an initial population of 60 *Tenebrio molitor*, a type of beetle, into a habitat and monitored the population over time. When the number of *Tenebrio molitor* in the habitat reached 160% of the initial population, the entomologist moved 75% of the *Tenebrio molitor* to a second habitat. How many *Tenebrio molitor* did the entomologist move to the second habitat at this time?

STUDENT-PRODUCED RESPONSE

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In the xy-plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 28x + 4$. If line s also passes through the point $(6, d)$, what is the value of d?

STUDENT-PRODUCED RESPONSE

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$$12(w-a)(w+8)=4(w+8)$$

In the given equation, a is a constant. The sum of the solutions to the equation is $\frac{34}{3}$. What is the value of a?

STUDENT-PRODUCED RESPONSE

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Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 58 square centimeters (cm^2), and the surface area of right rectangular prism Y is 1,450 cm^2. The volume of right rectangular prism Y is 1,500 cubic centimeters (cm^3). What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

STUDENT-PRODUCED RESPONSE

SECTION THREE · QUESTIONS 21–30

BONUS ROUND

Multiple choice and student-produced response questions.

Two identical rectangular prisms each have a height of 90 centimeters (cm). The base of each prism is a square, and the surface area of each prism is K cm^2. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K$ cm^2. What is the side length, in cm, of each square base?

A) 4

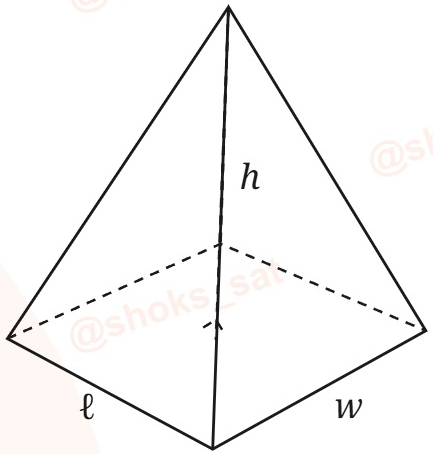
B) 8

C) 9

D) 16

The figure shown is a right rectangular pyramid, where $\ell = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

STUDENT-PRODUCED RESPONSE



Note: Figure not drawn to scale.

During the first 8.00 seconds after a car started moving, its speed increased to 11.0 meters per second. From 8.00 seconds to 14.0 seconds after the car started moving, its speed increased from 11.0 meters per second to 23.0 meters per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the car, in meters per second per second, during the first 8.00 seconds after the car started moving and the average rate of change of the speed of the car, in meters per second per second, from 8.00 seconds to 14.0 seconds after the car started moving?

STUDENT-PRODUCED RESPONSE

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$$x^2 + y^2 - 6x - 10y - 4n = 0$$

The given equation represents circle A in the *xy*-plane, where n is a constant. Point $(5, 7)$ lies on circle B, which has the same center but twice the diameter of circle A. What is the value of n?

STUDENT-PRODUCED RESPONSE

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A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 30% of the vehicles for sale were sedans and 50% were SUVs. If there were 57 sedans for sale at the dealership on Monday, how many minivans were for sale?

STUDENT-PRODUCED RESPONSE

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For a party, 65 hamburger buns are needed. Hamburger buns are sold in packages of 12. What is the minimum number of packages that should be bought for the party?

STUDENT-PRODUCED RESPONSE

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In triangle XYZ, the measure of angle X is x°, the measure of angle Y is y°, and the measure of angle Z is 31°. If $x > y > 31$, which of the following additional pieces of information is NOT sufficient to determine the values of x and y?

A) The value of $31 - x - 3y$

B) The value of $31 + 3x + y$

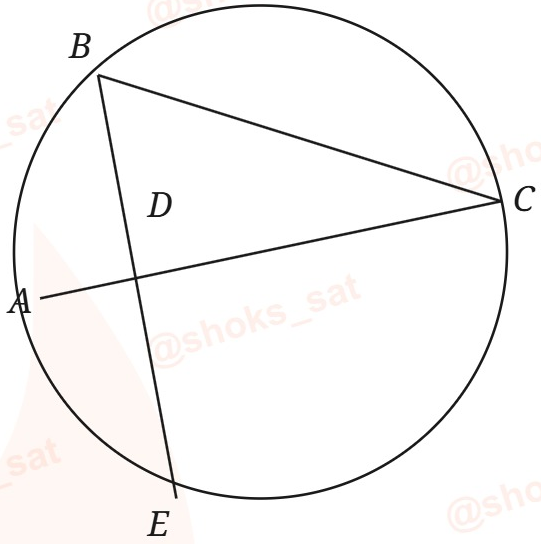
C) The value of $31 - 3x + 3y$

D) The value of $31 - 3x - 3y$

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In the figure shown, points A, B, C, and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D, and $BD = \sqrt{346}$. The diameter of the circle is 175. If $\frac{CD}{AD} = r$, what is the value of r?

STUDENT-PRODUCED RESPONSE



Note: Figure not drawn to scale.

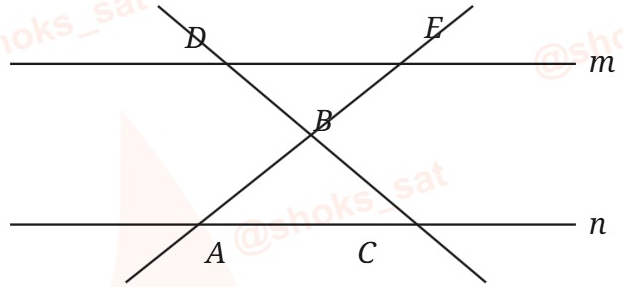
In the figure, line m is parallel to line n, and lines AE and CD intersect at point B. Which additional piece of information is sufficient to prove that triangle ABC is congruent to triangle EBD?

A) $AB = 16$ and $DB = 16$

B) $AB = 16$ and $EB = 16$

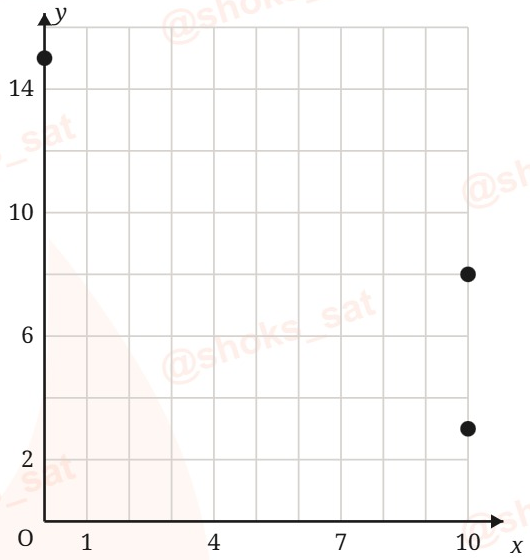
C) Triangles ABC and EBD are isosceles.

D) No additional information is necessary.



NOTE

The points $(0, 15)$, $(10, 8)$, and $(10, 3)$ are shown in the xy-plane, where the x-axis and y-axis are measured in units. These points define one of the bases of a triangular prism. The distance between the two bases of the prism is 20 units. What is the volume, in cubic units, of the prism?



STUDENT-PRODUCED RESPONSE

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COMPLETE ANSWER KEY · Q1–Q30

Q1	A) 5.8	**Q11**	C) $(x-9)^2+y^2=81$	**Q21**	B) 8
Q2	C) $y=13(x-1)+17$	**Q12**	C) 17	**Q22**	527.69
Q3	A) 23	**Q13**	C) 36	**Q23**	0.63
Q4	B) $4f+5s\geq 80$	**Q14**	12	**Q24**	-8
Q5	B) $\frac{53}{63}$	**Q15**	1,840	**Q25**	38
Q6	D) -68	**Q16**	-13	**Q26**	6
Q7	B) 17	**Q17**	72	**Q27**	D
Q8	D) 7,743	**Q18**	168	**Q28**	86.5
Q9	D) $7x^2-24x+144=0$	**Q19**	19	**Q29**	B
Q10	C) $x-4$	**Q20**	1,512	**Q30**	500

Good luck — you've got this!

With love, Shoks Team ❤️